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OBB Axis Semantics

(major vs minor)



Problem

For elongated objects (wine bottles, books, frying pans), grasping along the wrong object axis tips or drops the item. A generic top-down quat ignores object orientation.



When to Apply

- Elongated objects where grasp yaw matters
- Top-down grasps where gripper opening must align with a specific axis



Strategy

Compute the object's oriented bounding box (OBB); align yaw with the major axis for cylindrical grasps and minor axis for flat grasps. Geometry-derived, not hand-tuned.

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Container Disambig.

(cabinet vs wine rack)

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OBB Axis Semantics

```
obb = compute_obb(object_points)
major_axis = obb.axes[0] # Longest
minor_axis = obb.axes[1]

# Wine bottle: align along major axis
grasp_yaw = np.arctan2(major_axis[1], major_axis[0])

# Book: align along minor axis
grasp_yaw_book = np.arctan2(minor_axis[1],
                             minor_axis[0])
```



Scene Reasoning